

Distributor API Test Report v1 — Mouser + Digikey

Date: 2026-05-17 (first iteration of the API track)
Test parts: `BT168GW, 115` (WeEn SCR, SC-73), `STM32G030F6P6` (ST 32-bit MCU, TSSOP-20)
Stack: Python 3.10.9, `requests` 2.34.2, `python-dotenv` 1.2.2. No browser, no TLS impersonation — these are first-party REST APIs.

TL;DR

#	Source	Status	Endpoint	Auth	Quality
1	Mouser Search API v1	working	<code>POST /api/v1/search/partnumber</code> (fallback <code>/keyword</code>)	API-key in query string	high
2	Digikey Product Information API v4	working	<code>POST /products/v4/search/keyword</code>	OAuth2 client_credentials → bearer	high

Both APIs returned full canonical-schema records for both test parts on the very first call — no rate-limit retries, no CAPTCHAs, no `bm-verify` / Akamai walls. This is exactly what motivated the API track: Mouser is unreachable for scraping (blocked by Akamai BMP) but ships a free Search API; Digikey works for both but the API is one round-trip vs. ~30 s of Cloudflare wait under Playwright.

Output schema (shared with the scraper track)

Every per-variant record emits the same five fields the scraper track uses (see `scraper/doc/scraper_report_v2.md` and `common/_summary.py`):

Field	Meaning
<code>stock_now_qty</code>	现货 quantity — distributor's own warehouse
<code>stock_now_ship_text</code>	发货时间 string for 现货
<code>stock_future_qty</code>	期货/在途 quantity (<code>null</code> for unbounded factory order)
<code>stock_future_ship_text</code>	Lead-time string for the future pool
<code>stock_breakdown</code>	<code>[[label, warehouse, quantity, ship_text, note?], ...]</code> using the API's own field names

Site-native fields are preserved verbatim under `site_*` keys (`site_availability`, `site_factory_stock`, `site_lead_time`, `site_availability_on_order` for Mouser; `site_quantity_available`, `site_manufacturer_lead_weeks`, `site_normally_stocking`, `site_back_order_not_allowed`, `site_non_stock`, `site_discontinued`, `site_end_of_life`, `site_date_last_buy_chance`, `site_product_status` for Digikey).

MPN-variant grouping rule. A keyword search that returns multiple distinct `ManufacturerPartNumber` strings → one subfolder per variant. Never aggregate across MPN strings (see `feedback_mpn_variant_grouping.md`).

Folder layout (per run)

```
test/api_test/Test_<MPN>_<CHANNEL>_<YYYYMMDD>_<HH>_<MM>_<SS>/
├─ parent_summary.md           # Markdown overview + variant index
├─ <MPN>.json                  # Parent run record + variants_summary[]
├─ raw_response.json           # Full API payload (for audit)
├─ <variant_mpn>/              # One per distinct returned MPN string
│   ├─ <variant_mpn>.json       # Normalized canonical-schema record
│   ├─ <variant_mpn>_raw_part.json (Mouser) / _raw_product.json (Digikey)
│   └─ <variant_mpn>_summary.md # Rendered by common/_summary.py
```

Mirrors the scraper track's LCSC v3 / Future Electronics layout; the only differences are `test/api_test/` instead of `test/scraper_test/` and the per-variant raw file naming.

1. Mouser Search API v1 — working

Script: `api/scripts/api_mouser.py`

Base URL: `https://api.mouser.com/api/v1`

Auth: API key in query string (`?apiKey=<KEY>`). Free; register at <https://www.mouser.com/api-hub/>.

Primary endpoint: `POST /search/partnumber` with body

```
{"SearchByPartRequest": {"mouserPartNumber": "<MPN>", "partSearchOptions": "Exact"}}
```

Fallback endpoint: `POST /search/keyword` (used if partnumber search returns 0). Body:

```
{"SearchByKeywordRequest": {"keyword": "<MPN>", "records": 50, "startingRecord": 0,
                             "searchOptions": "", "searchWithYourSignUpLanguage": ""}}
```

Response shape (per part)

Each item in `SearchResults.Parts[]` carries:

- Identity: `ManufacturerPartNumber`, `MouserPartNumber`, `Manufacturer`, `Description`, `Category`, `LifecycleStatus`, `ROHSStatus`, `ProductDetailUrl`, `DataSheetUrl`, `ImagePath`.
- Stock: `Availability` (display string like "108590 库存量"), `AvailabilityInStock` (numeric string), `FactoryStock` (numeric string), `AvailabilityOnOrder[]` ({ `Quantity`, `Date` } pending PO batches), `LeadTime` (string).
- Ordering: `Min`, `Mult`, `PriceBreaks[]`. { `Quantity`, `Price`, `Currency` }.
- Specs: `ProductAttributes[]`. { `AttributeName`, `AttributeValue` }.
- Compliance: `ProductCompliance[]`. { `ComplianceName`, `ComplianceValue` } (HTS / ECCN / TARIC etc.).
- Misc: `AlternatePackagings[]`. `APMfrPN` (e.g. TR variant of base MPN), `UnitweightKg`. `Unitweight`.

Canonical mapping

Canonical	Mouser source
stock_now_qty	AvailabilityInStock (numeric) — fallback: leading integer from Availability
stock_now_ship_text	"Mouser 在库, 下单后立即发货" (constant when in-stock)
stock_future_qty	FactoryStock if > 0; otherwise null (unbounded factory order) when LeadTime is set
stock_future_ship_text	"原厂标准交货期 <LeadTime>"
stock_breakdown rows	Availability → in-stock row; FactoryStock + LeadTime → factory-stock row; each AvailabilityOnOrder[] entry → its own onorder row with the expected arrival date

Test results

Query	Variants	Highlight
STM32G030F6P6	1 (STM32G030F6P6 , Mouser P/N 511-STM32G030F6P6)	现货 108,590 + FactoryStock 0 (LeadTime 210 天数 → unbounded factory order); 8 price tiers ¥9.00 → ¥4.93
BT168GW,115	1 (BT168GW,115 , Mouser P/N 771-BT168GW-T/R)	现货 0 + on-order 13,694 arriving 2026-05-26 (LeadTime 112 天数); 9 tiers ¥4.83 → ¥0.864

Runs: test/api_test/Test_STM32G030F6P6_MOUSER_20260517_12_49_48/ and test/api_test/Test_BT168GW_115_MOUSER_20260517_12_50_52/ .

Notes / gotchas

- API key is locale-bound.** This key was registered on Mouser China — every response comes back localized to zh-CN: Availability is "108590 库存量", parameter names are "封装" / "标准包装数量", currency is RMB (¥), and LeadTime is in **days** suffixed with "天数" (NOT weeks). When the buyer asks for "lead time in weeks", we'd need a different key registered on the US site, or a post-processing conversion.
- Availability **mixes value + unit.** Parse the leading integer (re.match(r"-?\d+")) — don't compare to literal strings. Sometimes AvailabilityInStock carries the bare number; prefer it when present.
- AvailabilityOnOrder **is a real signal.** For BT168GW,115, Mouser had 0 in-stock but 13,694 already-ordered-from-factory units arriving 2026-05-26. This is buyer-relevant — surfaced as its own onorder row so it doesn't get hidden inside the abstract "factory order" pool.
- partSearchOptions: "Exact" **is intolerant of separators.** A query containing commas / spaces matched fine in our two test parts, but ambiguous keywords can fall through to 0 results — the fallback /search/keyword covers those. Both attempts are recorded in attempts .
- No OAuth, no token refresh, no throttle yet.** Free tier allows 1000 calls/day. For batch jobs we'll want

to add a simple per-second sleep and 429-retry-with-backoff.

2. Digikey Product Information API v4 — working

Script: `api/scripts/api_digikey.py`

Base URL: `https://api.digikey.com`

Auth: OAuth2 `client_credentials` flow. Register at <https://developer.digikey.com/> → create a Production app → get `DIGIKEY_CLIENT_ID` + `DIGIKEY_CLIENT_SECRET`.

Token endpoint: `POST /v1/oauth2/token` with form body `client_id=...&client_secret=...&grant_type=client_credentials` → `{access_token, expires_in: 599, token_type: "Bearer"}`.

Search endpoint: `POST /products/v4/search/keyword` with:

- Headers: `Authorization: Bearer <token>`, `X-DIGIKEY-Client-Id: <id>`, `X-DIGIKEY-Locale-Site: US`, `X-DIGIKEY-Locale-Language: en`, `X-DIGIKEY-Locale-Currency: USD`, `X-DIGIKEY-Customer-Id: 0`.
- Body: `{"Keywords": "<MPN>", "Limit": 50, "Offset": 0}`.

Response shape

Two parallel arrays — `ExactMatches[]` (MPN matches exactly) and `Products[]` (broader keyword match). We dedupe by `ManufacturerProductNumber` and prefer exact-match order.

Each Product carries:

- Identity: `ManufacturerProductNumber`, `Manufacturer.Name`, `Description.{ProductDescription, DetailedDescription}`, `Category.Name`, `DatasheetUrl`, `PhotoUrl`, `ProductUrl`, `ProductStatus.Status` (e.g. "Active").
- Stock: `QuantityAvailable` (top-level, the DK warehouse inventory total), `ManufacturerLeadWeeks` (string like "30" — bare integer, **assumed weeks**), `NormallyStocking` (bool), `BackOrderNotAllowed` (bool), `NonStock`, `Discontinued`, `EndOfLife`, `DateLastBuyChance`.
- Variations: `ProductVariations[]` — one per packaging form. Each has `DigikeyProductNumber`, `PackageType.Name` (Tube / Tape & Reel (TR) / Cut Tape (CT) / Digi-Reel®), `QuantityAvailableforPackageType`, `StandardPricing[].{BreakQuantity, UnitPrice, TotalPrice}`, `MinimumOrderQuantity`, `StandardPackage`.
- Specs: `Parameters[].{ParameterText, valueText}` (24 specs for STM32G030F6P6, 14 for BT168GW,115).
- Classifications (CORRECTED keys for v4): `Classifications.HtsusCode` (US HTS), `Classifications.ExportControlClassNumber` (ECCN — sometimes "EAR99"), `Classifications.RohsStatus`, `Classifications.ReachStatus`, `Classifications.MoistureSensitivityLevel`. **Not** `HtsCode` / `EccnCode` — those don't exist on v4.

Canonical mapping

Canonical	Digikey source
stock_now_qty	QuantityAvailable
stock_now_ship_text	"下单后立即发货" (constant when stock > 0)
stock_future_qty	null (unbounded — factory order with no committed quantity) when NormallyStocking is true or ManufacturerLeadWeeks is set
stock_future_ship_text	"原厂标准交货期 <N> weeks"
stock_breakdown rows	One QuantityAvailable row, one Factory Stock (ManufacturerLeadWeeks) row, plus one Packaging – <PackageType> row per ProductVariations[] entry (so the buyer can see the tube vs. reel split inside the DK warehouse total)

Test results

Query	Variants captured (exact + fuzzy)	Highlight
STM32G030F6P6	6 (1 exact + 5 fuzzy) — STM32G030F6P6 (497-STM32G030F6P6-ND, 现货 76,636, 30 weeks lead, 9 tiers \$1.58→\$0.766), STM32G030F6P6TR (497-STM32G030F6P6TR-ND, 现货 12,073), plus 4 unrelated keyword matches (SM-I-010 , U024-V2 , M137 , SM-I-005)	Exact match returns the base MPN; the TR variant is also surfaced (separate folder). 4 stale keyword matches are noise (kept as variants so we don't accidentally drop a real alternative)
BT168GW,115	2 — BT168GW,115 (1740-1084-2-ND, 现货 4,424, 26 weeks lead, 6 tiers \$0.178→\$0.130) + BT168GWF,115 (1740-1085-2-ND, 现货 37,597)	The base part and the "F" variant are clearly distinct SKUs — kept separate per the MPN-variant rule

Runs: test/api_test/Test_STM32G030F6P6_DIGIKEY_20260517_12_56_11/ and test/api_test/Test_BT168GW_115_DIGIKEY_20260517_12_56_36/.

Notes / gotchas

- Lead time is a bare integer.** ManufacturerLeadWeeks: "30" — no unit suffix. The convention is **weeks** (verified against digikey.cn's "标准交货期 30 周" rendering). Normalizer appends " weeks" in the summary; raw site_manufacturer_lead_weeks keeps the bare integer for audit.
- Classification keys are different from v3.** v4 uses HtsusCode (not HtsCode), ExportControlClassNumber (not EccnCode), and exposes ReachStatus + MoistureSensitivityLevel. Adjusted the normalizer to read these correctly; fallback to v3 names kept in case the API drift goes the other way.
- Keyword search returns noise.** A keyword like STM32G030F6P6 returned 4 unrelated matches (relay

boards, sensor kits) alongside the actual MPN. We keep them all as separate variants per the MPN-variant rule, but a future enhancement could score-rank by string similarity and demote obvious misfires (or use `productdetails` endpoint when we already know the canonical DK P/N).

4. **OAuth token caches 599 s.** We don't currently cache the token across runs — every script invocation pays one token round-trip (~150 ms). For high-volume batch runs we should cache it to a file (`.token_cache.json` with expiry timestamp).
5. **Locale picks currency.** Setting `X-DIGIKEY-Local-Currency: USD` gives USD pricing; `CNY` would give CNY. We use USD for cross-batch consistency since the scraper-track Digikey scrape also reads from `digikey.cn` → CNY isn't apples-to-apples anyway.
6. `isBackOrderAllowed` **lives off** `BackOrderNotAllowed`. v4 uses the inverted name — `BackOrderNotAllowed: false` means back-order IS allowed. Read carefully.

What this enables

- **Mouser:** previously a complete dead end (Akamai bm-verify wall in the scraper track) — now fully working with high data quality, including the on-order batch detail that the website itself displays.
- **Digikey:** previously needed ~30 s of Playwright + Cloudflare clear per part — now one OAuth round-trip + one search call (~500 ms total). For batch jobs across 100s of MPNs, this is two orders of magnitude faster.
- **Cross-track comparability:** both APIs map onto the same canonical schema as `scrape_1csc_v3.py` / `scrape_digikey.py` / `scrape_hqew.py` / `scrape_future.py`. The same `common/_summary.py` renders all of them. A buyer reading `<MPN>_summary.md` cannot tell whether the data came from scraping or an API — by design.

Auth + secrets handling

- API keys live in `api/.env` (gitignored). Scripts read via `python-dotenv` → `os.environ.get(...)` only — never write the keys to logs or output files.
- `api/.env.example` documents the env-var names without values.
- The `attempts[]` log records `status` / `len` / `outcome` per call but never includes request bodies (which carry credentials) or response bodies past 500 chars (Mouser auth errors come through in plain text; truncated to avoid leaking).

Future work / backlog

- **Octopart / Nexar** — aggregator that covers Arrow (also scraper-blocked by Akamai). Free tier available via Nexar developer portal.
- **Element14 (Farnell) API** — another distributor for EU coverage.
- **Token cache for Digikey** — file-based or in-process; 599 s TTL.
- **429 / rate-limit handling** — exponential backoff + per-second throttling for batch sweeps.
- **Sandbox URLs** — Digikey has `sandbox-api.digikey.com`; useful for development without burning the daily quota. Not yet implemented.
- **Currency conversion** — Mouser key returns RMB, Digikey returns USD. For BOM-level price

comparison, we'll want a single canonical currency (USD likely) with FX-rate metadata stamped on the run.

- **De-noising Digikey keyword results** — when 4 of 6 returned variants are unrelated to the queried MPN, the variants table gets crowded. Score by `manufacturer_part_number` prefix-similarity and surface the noise in a separate "unrelated keyword matches" section of the parent summary.

Files of record

- `api/scripts/api_mouser.py` — Mouser Search API v1 client.
- `api/scripts/api_digikey.py` — Digikey Product Information API v4 client.
- `api/requirements.txt` — `requests`, `python-dotenv` (no Playwright / `curl_cffi` needed).
- `api/.env.example` — API-key placeholders. Real `.env` is gitignored.
- `api/README.md` — track-level overview + conventions.
- `common/_summary.py` — shared per-MPN summary renderer (scraper + API).
- Memory: `MEMORY.md` indexes the cross-track feedback rules (`feedback_test_output_folder.md`, `feedback_stock_breakdown_fields.md`, `feedback_mpn_variant_grouping.md`, `feedback_site_native_fields.md`) — all apply equally to this track.
- Counterpart scraper-track report: `scraper/doc/scraper_report_v2.md`.